TechCorp Solutions — System Architecture & Technical Specifications

1. Architecture Overview

TechCorp's platform is built on a microservices architecture deployed on AWS. The system processes over 2 million API requests per day across 12 microservices, with 99.97% uptime SLA. The architecture follows domain-driven design principles with event-driven communication via Apache Kafka.

Core architectural decisions: • Polyglot persistence: PostgreSQL (transactional), MongoDB (documents), Redis (cache), Elasticsearch (search) • API Gateway: Kong for rate limiting, auth, and routing • Service mesh: Istio for inter-service communication and observability • CI/CD: GitHub Actions → ArgoCD → Kubernetes (EKS)

2. Database Schema

Primary PostgreSQL schema (techcorp_prod database):

- users: id, email, name, role, org_id, created_at, last_login, is_active
- organisations: id, name, plan, seats, created_at, billing_email, stripe_customer_id
- documents: id, org_id, filename, file_type, size_bytes, chunk_count, indexed_at, status
- conversations: id, session_id, user_id, created_at, message_count, last_message_at
- messages: id, conversation_id, role, content, tokens_used, response_time_ms, created_at
- api_keys: id, org_id, key_hash, name, permissions, created_at, expires_at, last_used_at
- usage_logs: id, org_id, endpoint, tokens, cost_usd, latency_ms, status_code, created_at

3. API Endpoints

REST API v2 (base: api.techcorp.com/v2). Authentication: Bearer JWT token.

- POST /auth/login — Returns JWT access token (expires 24h)
- POST /auth/refresh — Refresh expired token using refresh token
- GET /documents — List documents with pagination (?page=1&limit;=20)
- POST /documents/upload — Upload document (multipart/form-data, max 50MB)
- DELETE /documents/{id} — Delete document and remove from index
- POST /chat — Send question and receive streamed answer (SSE)
- GET /chat/history/{session_id} — Get conversation history

- GET /usage — Get token usage statistics for billing period
- GET /health — System health check (public endpoint, no auth required)

4. Security Architecture

TechCorp implements a defence-in-depth security model aligned with ISO 27001 and SOC 2 Type II standards.

- Authentication: OAuth 2.0 + PKCE for user auth, API keys for service-to-service
- Authorisation: Role-Based Access Control (RBAC) with 5 roles: Owner, Admin, Developer, Viewer, Billing
- Encryption: TLS 1.3 in transit; AES-256 at rest for all stored data
- Secrets Management: AWS Secrets Manager for all credentials and API keys
- Network: VPC with private subnets; EC2 instances not publicly accessible
- WAF: AWS WAF with OWASP Core Rule Set blocking common attacks
- Penetration Testing: Quarterly by certified external security firm
- VAPT: Automated daily scans via Snyk (dependencies) and SonarQube (code)
- Audit Logs: All API calls logged to immutable S3 with 7-year retention

5. Deployment Guide

Production deployment uses Kubernetes on AWS EKS (us-east-1 primary, ap-south-1 secondary for disaster recovery).

Prerequisites: kubectl 1.28+, helm 3.12+, AWS CLI 2.x, access to techcorp-prod cluster.

Deploy a service: helm upgrade --install documind-api ./helm/documind-api \ --namespace production \ --set image.tag=$GIT_SHA \ --set replicaCount=3 \ --values helm/values.prod.yaml

Rolling update strategy: maxSurge=1, maxUnavailable=0 (zero-downtime deploys). Canary deployments: 10% traffic → 50% → 100% with automated rollback on error rate >1%.

6. Monitoring & Observability

- Metrics: Prometheus + Grafana dashboards (grafana.techcorp.internal)
- Logs: ELK Stack (Elasticsearch, Logstash, Kibana) — 30-day retention
- Tracing: Jaeger distributed tracing (sampled at 10% in production)
- Alerts: PagerDuty for P0/P1 incidents; Slack #alerts for P2/P3

- SLOs: 99.97% availability, p99 latency <500ms, error rate <0.1%
- On-call rotation: Engineering teams on 1-week rotation, 15-min response SLA
- Incident runbooks: Stored in Confluence at docs.techcorp.internal/runbooks